# Esame 18/7/25

Es. 1 Diagrammi di Bode e Nyquist

$$F(s) = \frac{k(s-1)}{(s+10)(s^2+s+1)} \qquad k > 0$$

2. Forma di Bode

$$F(s) = -\frac{k}{10} \cdot \frac{1-s}{(1+s/10)(1+s+s^2)}$$

$k > 0 \Rightarrow$ il "−" va lasciato.

TERMINE TRINOMIO

$\omega_n = 1$

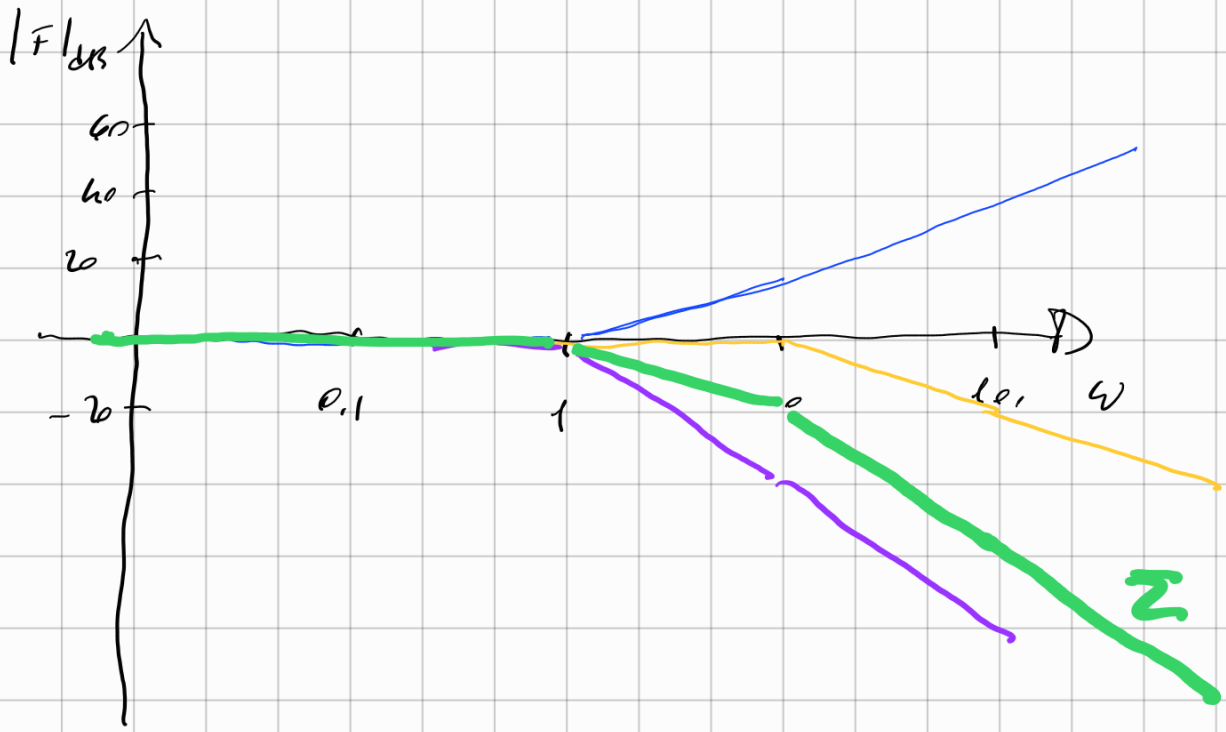
LIMITE PER USARE IL DIAG ASINTOTICO X MODULO $\rightarrow \xi = 1/2$

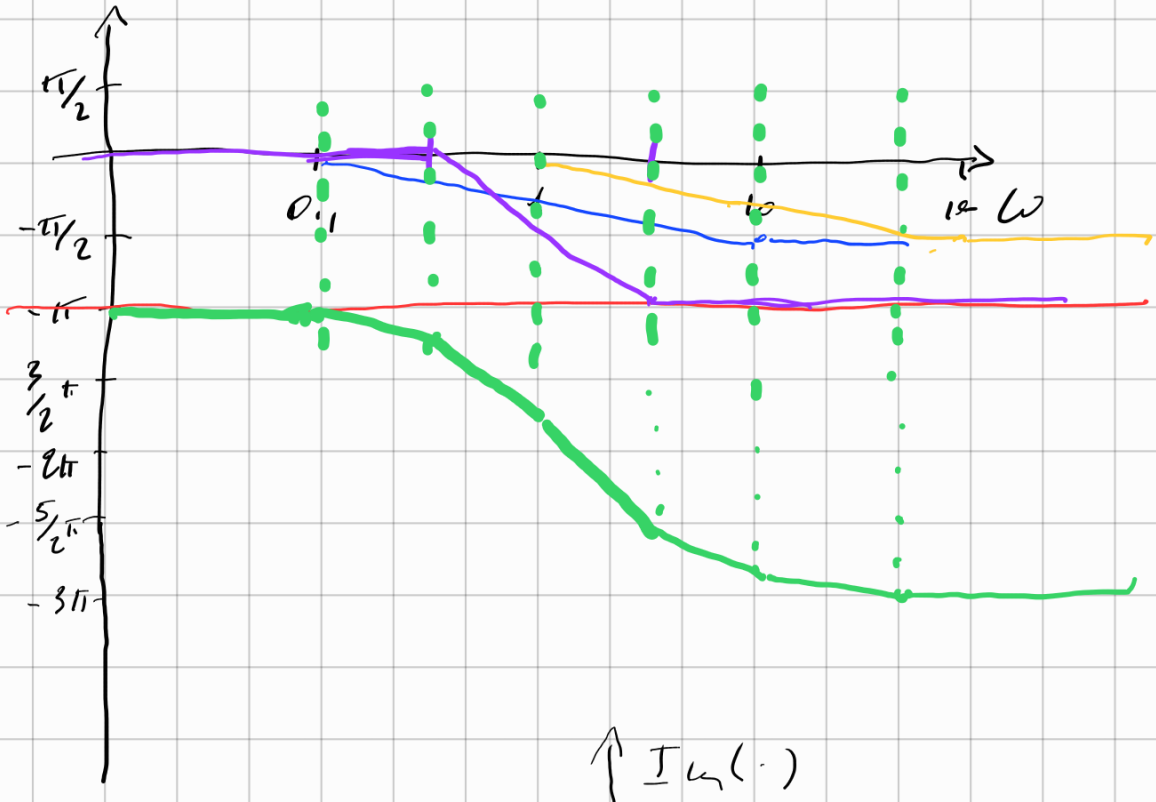
$k = 10$ o $k = 1$

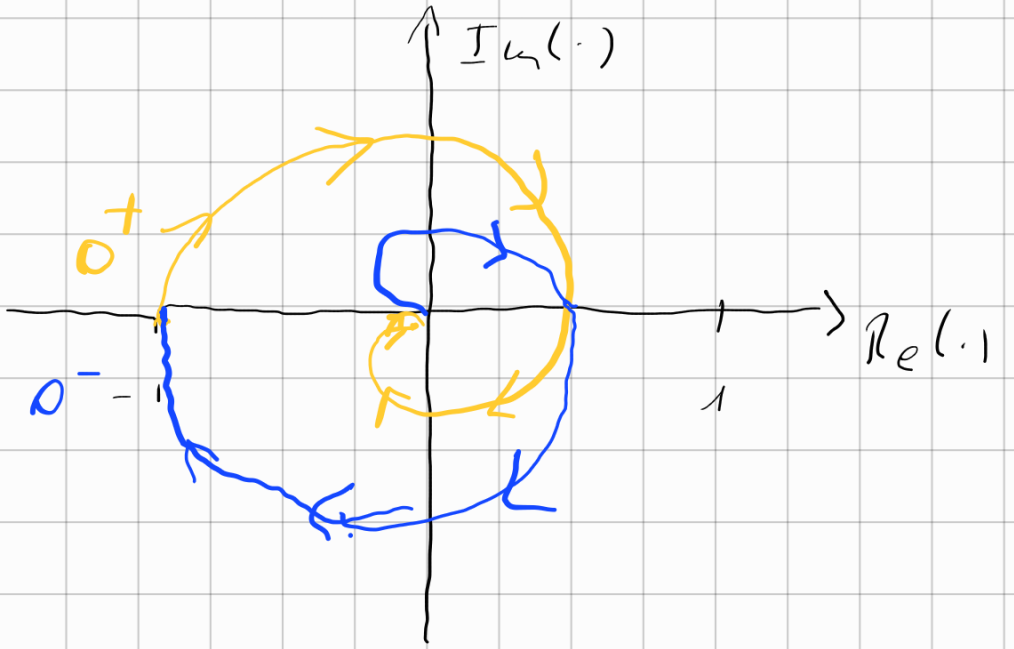
solo uno più "sega"

$k = 10$

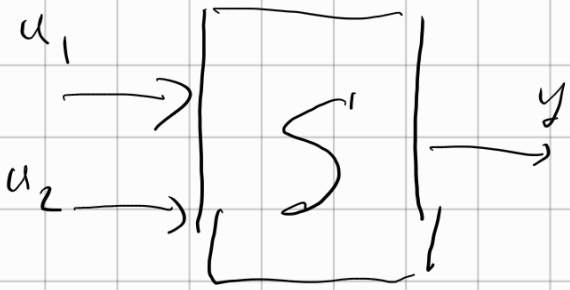
5. TRACCIAMENTO DEI SINGOLI TERMINI !!!


|F|dB
60
40
20
-20
0.1
1
10
ω
-2


π/2
-π/2
-π
3/2 π
-2π
-5/2π
-3π
0.1
1
10
ω


Im(·)
Re(·)
0+
0-
-1
1

Es. 2




$$y(z) = \begin{pmatrix} W_1(z) & W_2(z) \end{pmatrix} u(z) =$$

$$y(z) = \begin{pmatrix} W_1(z) & W_2(z) \end{pmatrix} \begin{pmatrix} u_1(z) \\ 0 \end{pmatrix} =$$

$$= W_1(z) u_1(z)$$

UGUALE PER $u_2$

$$W_1(z) = \frac{y_1(z)}{u_1(z)}\Big|_{u_2=0} \qquad W_2(z) = \frac{y_2(z)}{u_2(z)}\Big|_{u_1=0}$$

$$u_1(z) = \sum_{t=0}^{\infty} u_1(t) z^{-t} = 1 + \frac{1}{z} + \frac{1}{z^2} = \frac{z^2+z+1}{z^2}$$

$$u_2(z) = \mathcal{Z}\left[\delta_{-1}(t)\right] = \frac{z}{z-1}$$

$$y_1(z) = \frac{1}{z^2} + \frac{2}{z^3} + \frac{1}{z^4} = z^{-1}\left(\frac{1}{z} + \frac{2}{z^2} + \frac{1}{z^3}\right)$$

↑ RITARDO (IN $\phi$ NON C'È NULLA)

$$y_2(z) = \mathcal{Z}\left[\delta_{-1}(t-1) - \delta(t-1)\right] =$$

$$= z^{-1}\left(\frac{z}{z-1} - 1\right)$$

$$W_1(z) = \frac{\frac{(z^2+2z+1)}{z^4}}{\frac{z^2+z+1}{z^2}} = \frac{z^2+2z+1}{(z^2+z+1)z^2}$$

$$W_2(z) = \frac{z^{-1}\left(\frac{z}{z-1} - 1\right)}{\left(\frac{z}{z-1}\right)} = z^{-1}\,\frac{\frac{1}{z-1}}{\frac{z}{z-1}} = \frac{1}{z^2}$$

$$W(z) = \begin{pmatrix} \dfrac{z^2+2z+1}{z^2(z^2+z+1)} & \dfrac{1}{z^2} \end{pmatrix}$$

Forma canonica osservabile

$$W(s) = \frac{\begin{pmatrix} z^2+2z+1 & z^2+z+1 \end{pmatrix}}{z^2(z^2+z+1)} =$$

$$= \frac{(1\ \ 1)z^2 + (2\ \ 1)z + (1\ \ 1)}{z^2(z^2+z+1)}$$

$$A_0 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{pmatrix} \qquad B_0 = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$$

$$C_0 = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}$$

Es. 3.2

$$F(z) = Z\left[\mathcal{L}^{-1}\left[F(s)\cdot\frac{1}{s}\right]_{t=kT}\right]\frac{z-1}{z}$$

$$\mathcal{L}^{-1}\left[F(s)\cdot\frac{1}{s}\right] = \mathcal{L}^{-1}\left[\frac{-\frac{1}{10}}{s} + \frac{-\frac{11}{90}}{s+10} + \frac{\frac{2}{9}}{s+1}\right]$$

$$= -\frac{1}{10} - \frac{11}{90}e^{-10t} + \frac{2}{9}e^{-t}$$

$$\mathcal{L}^{-1}\left[F(s)\cdot\frac{1}{s}\right]_{t=kT} = -\frac{1}{10} - \frac{11}{90}e^{-10Tk} + \frac{2}{9}e^{-Tk}$$

$$F(z) = \mathcal{Z}\left[\mathcal{L}^{-1}\left[F(s)\frac{1}{s}\right]_{t=kT}\right]\cdot\frac{z-1}{z} =$$

$$= -\frac{1}{10} - \frac{11}{90}\frac{z-1}{z-e^{-10T}} + \frac{2}{9}\frac{z-1}{z-e^{-T}}\Bigg|_{T=0.5} =$$

$$= -\frac{1}{10} - \frac{11}{90}\frac{z-1}{z-e^{-5}} + \frac{2}{9}\frac{z-1}{z-e^{-0.5}}$$

Es. 3b

Sol. 1: Avendo la F.d.T., i modi vari sono solo quelli eccitabili ed osservabili;

Sol. 2, Sol. 3, ...

Tutte comportano calcoli inutili per arrivare a risoluzione, direttamente o indirettamente, la Sol 1.